

SOP (Standard Operating Procedure) for RPO & RTO–driven Disaster Recovery in SQL Server, aligned to SQL Server 2016–2025, covering On-Premises, AWS, and Azure, with real-time OLTP and OLAP scenarios and reflecting current enterprise trends.

SOP: Disaster Recovery Strategy for SQL Server (2016–2025)

Scope: On-Premises, AWS, Azure

Workloads: OLTP & OLAP

Focus: RPO / RTO compliance, high availability, resilience, and recoverability

1. Purpose

To define standardized procedures for designing, implementing, operating, and validating SQL Server Disaster Recovery (DR) solutions based on **Recovery Point Objective (RPO)** and **Recovery Time Objective (RTO)** for mission-critical workloads.

2. Applicability Matrix

Environment	SQL Versions	Hosting Models
On-Prem	2016–2022	Physical, VMware, Hyper-V
Azure	2016–2025	Azure VM, Azure SQL MI
AWS	2016–2025	EC2 + EBS, FSx
Workloads	All	OLTP, OLAP, Hybrid

3. RPO / RTO Classification (Standardized)

Tier	Business Criticality	RPO	RTO
Tier 0	Financial / Core OLTP	0–5 min	< 15 min
Tier 1	Revenue Impacting	≤ 15 min	≤ 1 hr
Tier 2	Operational	≤ 1 hr	≤ 4 hrs
Tier 3	Reporting / Archive	≥ 24 hrs	≥ 24 hrs

4. Technology Mapping (Latest Trend)

4.1 DR Technology Selection Matrix

RPO	RTO	Recommended Technology
Near-Zero	Seconds	Always On AG (Sync)
≤ 15 min	< 1 hr	AG (Async), Log Shipping
≤ 1 hr	≤ 4 hrs	Log Shipping
≥ 24 hrs	≥ 24 hrs	Backup + Restore

5. SOP by Workload Type

This section defines **standard operating procedures based on workload characteristics**, ensuring that **RPO/RTO, architecture, and operational actions** are aligned to how SQL Server is actually used (OLTP vs OLAP).

5.1 Workload Classification SOP

Step 1: Identify Workload Type

Criteria	OLTP	OLAP
Transaction Volume	High	Low–Moderate
Write Frequency	Very High	Low
Query Type	Short, frequent	Long-running

Criteria	OLTP	OLAP
Data Volatility	Constant	Mostly static
Business Impact	Immediate	Delayed

Action

- Classify each database as **OLTP**, **OLAP**, or **Hybrid**
- Document classification in CMDB / DR register

5.2 RPO / RTO Definition SOP**Step 2: Assign RPO & RTO Targets**

Workload	Typical RPO	Typical RTO
Tier-0 OLTP	0–5 minutes	< 15 minutes
Tier-1 OLTP	≤ 15 minutes	≤ 1 hour
Hybrid (OLTP + Reporting)	≤ 30 minutes	≤ 2 hours
OLAP / DW	1–24 hours	4–24 hours

Action

- Obtain **business sign-off** on RPO/RTO
- RPO/RTO must be **measurable and testable**

5.3 DR Technology Selection SOP**Step 3: Select DR Technology Based on Workload**

Workload	Primary DR Technology	Secondary
Mission-Critical OLTP	Always On AG (Sync)	Backup to Cloud
Business-Critical OLTP	AG (Async)	Log Shipping
Hybrid	AG + Read Replica	Log Shipping
OLAP	Log Shipping	Backup/Restore

Decision Rule

- If **RPO < 15 min** → **Log-based replication mandatory**
- If **RTO < 1 hr** → **Automated failover required**

5.4 Platform-Specific SOP (On-Prem / Cloud)**Step 4: Platform Mapping**

Platform	OLTP SOP	OLAP SOP
On-Prem	WSFC + AG	Backup + Restore
Azure	AG / Failover Groups	Geo-Restore
AWS	AG / Log Shipping	S3 Backup Restore

Action

- Ensure **network latency**, **storage IOPS**, and **licensing** support selected DR tech

5.5 Backup & Data Protection SOP**Step 5: Backup Strategy per Workload**

Workload	Full	Diff	Log
OLTP	Daily	4–6 hrs	5–10 min
Hybrid	Daily	6 hrs	10–15 min

Workload	Full	Diff	Log
OLAP	Daily	Optional	Optional

Mandatory Controls

- Backups must be:
 - Encrypted
 - Off-host
 - Cross-region / off-site
- Restore tests must be documented

5.6 Failover & Recovery SOP

Step 6: Recovery Procedure

OLTP

1. Detect failure (monitoring alert)
2. Trigger automatic or manual failover
3. Validate application connectivity
4. Confirm RPO/RTO compliance
5. Notify stakeholders

OLAP

1. Validate last successful backup
2. Restore to DR instance
3. Apply differential/logs if required
4. Validate reports
5. Resume scheduled jobs

5.7 DR Testing & Validation SOP

Step 7: Testing Frequency

Workload	Test Type	Frequency
Tier-0 OLTP	AG Failover	Quarterly
Tier-1 OLTP	Partial DR	Bi-annual
OLAP	Restore Test	Annual

Action

- Capture:
 - Actual RPO achieved
 - Actual RTO achieved
 - Gaps and remediation plan

5.8 Documentation & Ownership SOP

Step 8: Governance

Role	Responsibility
DBA	DR configuration & testing
Infra / Cloud	Platform availability
App Owner	Business validation
Security	Compliance & encryption

This section is the **control layer** that ensures:

- RPO/RTO are business-aligned

- DR choices are justified
- Audits and compliance reviews pass

6. OLTP Real-Time Scenario (Mission-Critical)

6.1 On-Premises (SQL Server 2019/2022)

Use Case: Banking / ERP / Order Processing

Target: RPO = 0–5 min, RTO < 15 min

Architecture

- Windows Server Failover Cluster (WSFC)
- Always On Availability Groups (Synchronous Commit)
- Automatic failover
- SAN or S2D storage

Operational SOP

1. Enable Full Recovery Model
2. Configure synchronous AG replicas
3. Enable automatic failover
4. Monitor AG latency (< 5 sec)
5. Run log backups every 5 minutes
6. Quarterly DR failover testing

Failure Scenario

- Primary node crash
- Automatic failover triggered
- Application reconnect via listener
- Data loss = 0

6.2 Azure OLTP (SQL 2022–2025)

Use Case: SaaS / Cloud-Native Apps

Target: RPO = 0–5 min, RTO < 30 min

Architecture

- Azure VM with Always On AG
- Availability Zones (AZ-aware)
- Azure Load Balancer
- Azure Backup (secondary)

Operational SOP

1. Deploy SQL VMs across AZs
2. Configure AG with zone redundancy
3. Enable Accelerated Networking
4. Configure automated failover
5. Test failover every 6 months

Failure Scenario

- Zone outage
- Automatic cross-zone failover
- Application resumes via AG listener

6.3 AWS OLTP (SQL 2019–2025)

Use Case: E-commerce / Payments

Target: RPO ≤ 15 min, RTO ≤ 1 hr

Architecture

- EC2 Multi-AZ
- Always On AG (Async)

- EBS io2 / FSx for Windows
- Route 53 for DNS failover

Operational SOP

1. Enable Multi-AZ deployment
2. Configure async AG replicas
3. Log backups every 5–10 min to S3
4. Snapshot EBS volumes
5. Monthly DR validation

Failure Scenario

- Primary EC2 instance failure
- Manual or scripted failover
- DNS redirect via Route 53

7. OLAP / Analytics Scenario

7.1 On-Prem OLAP (SQL 2016–2022)

Use Case: Data Warehouse / Reporting

Target: RPO = 24 hrs, RTO = 24 hrs

Architecture

- Standalone SQL
- Daily full + diff backups
- Offsite backup replication

Operational SOP

1. Full backup nightly
2. Differential backup every 6 hrs
3. Weekly restore testing
4. Archive backups to secondary site

7.2 Azure OLAP (SQL 2022–2025)

Use Case: Enterprise BI

Target: RPO = 1 hr, RTO = 2 hrs

Architecture

- Azure SQL Managed Instance
- Built-in automated backups
- Geo-restore or Failover Groups

Operational SOP

1. Enable Geo-replication
2. Validate restore points
3. Automate geo-restore scripts
4. Monitor backup retention

7.3 AWS OLAP (SQL 2019–2025)

Use Case: Cloud DW

Target: RPO = 1–4 hrs, RTO = 4 hrs

Architecture

- EC2 SQL Server
- Log Shipping to DR region
- S3 lifecycle-managed backups

Operational SOP

1. Schedule log shipping every 15–30 min
2. Monitor copy/restore latency

3. Quarterly regional DR test
4. Use automation (SSM / PowerShell)

8. Backup Strategy SOP (All Platforms)

Backup Type	Frequency	Purpose
Full	Daily	Base restore
Differential	4–6 hrs	Faster recovery
Log	5–15 min	RPO compliance

Best Practices

- Encrypt backups
- Store off-box and cross-region
- Test restores quarterly
- Monitor job failures

9. Monitoring & Validation

Mandatory KPIs

- Backup success rate = 100%
- AG latency < RPO threshold
- Restore time < RTO target
- DR test success = Pass

Tools

- SQL Agent Alerts
- Azure Monitor / Log Analytics
- AWS CloudWatch
- Third-party monitoring (optional)

10. DR Testing SOP

Test Type	Frequency
Backup Restore	Monthly
AG Failover	Quarterly
Full DR Simulation	Yearly

Documentation

- Capture RPO/RTO achieved
- Identify gaps
- Update architecture if breached

11. Key Trends (2024–2025)

- Increased use of **Availability Zones**
- Shift from Log Shipping to **Async AG**
- Automation using **PowerShell + Terraform**
- Hybrid DR (On-Prem → Cloud)
- Cost-optimized OLAP via backup-only DR

12. Ownership & Review

- **DBA Team:** Implementation & monitoring
- **Infra/Cloud Team:** Platform readiness
- **Business Owners:** RPO/RTO sign-off
- **Review Cycle:** Every 12 months or major change

SOP view of RPO/RTO–driven SQL Server Disaster Recovery for Banking, Healthcare, and Retail, aligned to SQL Server 2016–2025, On-Prem + AWS + Azure, and OLTP / OLAP workloads.

This is written in an **audit-ready, industry-practical format.**

Industry-Tailored SOP: SQL Server DR (RPO & RTO)

1. Banking Industry SOP

(Core Banking, Payments, Trading, Risk, Regulatory Reporting)

1.1 Business & Regulatory Context

- Zero or near-zero tolerance for data loss
- Regulatory oversight: PCI-DSS, SOX, ISO 22301, regional banking regulators
- Downtime = financial loss + regulatory penalties

1.2 Workload Classification

Workload	Type	Criticality
Core Transactions	OLTP	Tier-0
Payments / Clearing	OLTP	Tier-0
Risk & Fraud	Hybrid	Tier-1
Regulatory Reports	OLAP	Tier-2

1.3 RPO / RTO Targets (Banking)

Workload	RPO	RTO
Core OLTP	0–1 min	< 15 min
Payments	≤ 5 min	< 30 min
Risk Systems	≤ 15 min	≤ 1 hr
Reporting	≤ 24 hrs	≤ 24 hrs

1.4 Approved DR Architecture

On-Prem

- WSFC + Always On AG (Synchronous)
- Dedicated DR site
- Storage-level redundancy

Azure

- SQL Server on Azure VM
- Availability Zones + AG (Sync)
- Azure Load Balancer
- Azure Backup for compliance retention

AWS

- EC2 Multi-AZ
- Always On AG (Async cross-region)
- EBS io2 / FSx
- S3 Object Lock (immutable backups)

1.5 Operational SOP (Banking)

1. Log backups every **2–5 minutes**
2. AG health monitored continuously
3. Automatic failover for Tier-0 only

4. Manual approval required for cross-region DR
5. Quarterly regulator-observed DR test
6. Evidence retention for **7+ years**

1.6 Real Failure Scenario

Scenario: Primary DC power outage

Outcome:

- AG automatic failover in < 30 seconds
- No data loss
- Applications reconnect via listener
- Regulatory SLA met

2. Healthcare Industry SOP

(EHR, EMR, LIS, PACS, Patient Billing)

2.1 Business & Compliance Context

- Patient safety priority
- Compliance: HIPAA, HITECH, GDPR (EU)
- Data integrity > performance
- Downtime tolerated briefly but must be predictable

2.2 Workload Classification

Workload	Type	Criticality
Patient Records (EHR)	OLTP	Tier-0
Lab / Imaging	OLTP	Tier-1
Scheduling / Billing	OLTP	Tier-1
Analytics / Research	OLAP	Tier-2

2.3 RPO / RTO Targets (Healthcare)

Workload	RPO	RTO
EHR	≤ 5 min	≤ 30 min
Clinical Systems	≤ 15 min	≤ 1 hr
Billing	≤ 30 min	≤ 2 hrs
Research	≤ 24 hrs	≤ 24 hrs

2.4 Approved DR Architecture

On-Prem

- AG (Async preferred for distance)
- Manual failover with clinical approval
- Encrypted backups offsite

Azure

- Azure SQL Managed Instance
- Failover Groups
- Geo-restore
- Azure Key Vault for encryption keys

AWS

- EC2 SQL + Log Shipping

- Encrypted backups to S3
- Cross-region restore scripts

2.5 Operational SOP (Healthcare)

1. Encrypt data at rest and in transit
2. Log backups every **10–15 minutes**
3. Failover requires **clinical operations approval**
4. DR tests conducted **off-hours only**
5. Access logs retained for audit
6. Restore validation includes **data consistency checks**

2.6 Real Failure Scenario

Scenario: Ransomware attack on primary SQL host

Outcome:

- Primary isolated
- Restore to clean DR environment
- Last known good backup applied
- Minimal patient data loss within RPO

3. Retail Industry SOP

(POS, E-Commerce, Inventory, Promotions, BI)

3.1 Business Context

- Revenue directly tied to uptime
- High seasonal peaks (sales, holidays)
- Cost-optimized DR preferred
- Short outages acceptable if data preserved

3.2 Workload Classification

Workload	Type	Criticality
POS / Online Orders	OLTP	Tier-0
Inventory	OLTP	Tier-1
Promotions	OLTP	Tier-1
Sales Analytics	OLAP	Tier-2

3.3 RPO / RTO Targets (Retail)

Workload	RPO	RTO
POS / Orders	≤ 5–10 min	≤ 30 min
Inventory	≤ 30 min	≤ 2 hrs
Promotions	≤ 1 hr	≤ 4 hrs
Analytics	≤ 24 hrs	≤ 24 hrs

3.4 Approved DR Architecture

On-Prem

- AG (Async)
- Log Shipping for secondary DR
- Backup to cloud storage

Azure

- SQL VM with AG
- Azure Site Recovery (infrastructure)
- Auto-scale for seasonal load

AWS

- SQL on EC2
- Log Shipping to secondary region
- S3 + lifecycle policies

3.5 Operational SOP (Retail)

1. Increase log backup frequency during peak sales
2. Monitor replication lag continuously
3. Pre-approve failover during campaigns
4. Use read replicas for reporting
5. DR tests before major sale events

3.6 Real Failure Scenario

Scenario: Primary region outage during holiday sale

Outcome:

- Async AG / Log Shipping promoted
- DNS switch via Route 53 / Azure Traffic Manager
- Sales resume with minimal disruption

4. Cross-Industry Comparison Summary

Industry	Data Loss Tolerance	Automation Level	Compliance Pressure
Banking	Near-Zero	Very High	Extremely High
Healthcare	Very Low	Medium	High
Retail	Low–Moderate	High	Medium

5. Key Takeaway

RPO and RTO are not technical metrics — they are industry risk decisions.

- Banking optimizes for **data integrity**
- Healthcare optimizes for **patient safety**
- Retail optimizes for **revenue continuity**

The DR design, automation level, and testing rigor must reflect this reality.

A. INDUSTRY-SPECIFIC DR RUNBOOKS

(Operational, step-by-step, incident-ready)

A1. Banking DR Runbook (Tier-0 & Tier-1 Systems)**Trigger Conditions**

- Primary SQL Server unavailable > 60 seconds
- Data corruption detected
- DC or AZ outage
- Security incident impacting data integrity

Immediate Actions (T+0 to T+5 min)

1. Confirm incident via monitoring
2. Freeze application writes (if manual)
3. Notify:
 - DBA Lead
 - Infrastructure Lead
 - Business Continuity Officer
4. Validate AG health / last log received

Failover Execution

- **Tier-0**
 - Automatic AG failover (Sync)
- **Tier-1**
 - Manual failover after business approval

Post-Failover Validation

- Data consistency check
- Application connectivity validation
- Transaction reconciliation report

Compliance Actions

- Capture RPO/RTO achieved
- Retain logs and screenshots
- Incident report stored for audit (7+ years)

A2. Healthcare DR Runbook (Clinical Systems)**Trigger Conditions**

- EHR system unavailable
- Data integrity warning
- Cybersecurity incident (ransomware)

Immediate Actions

1. Notify Clinical Operations
2. Switch systems to read-only if required
3. Isolate affected infrastructure

Recovery Execution

- Restore from last known clean backup OR
- Promote DR replica (manual approval mandatory)

Validation

- Patient record integrity check
- Clinical workflow validation
- Audit access logs

Compliance

- HIPAA breach assessment
- Incident documentation retained per regulation

A3. Retail DR Runbook (Revenue Systems)**Trigger Conditions**

- POS / E-commerce outage
- Performance degradation during peak sales

Immediate Actions

1. Activate pre-approved DR window (if peak)
2. Notify business operations
3. Confirm replication lag

Recovery Execution

- Async AG failover OR
- Promote Log Shipping secondary

Validation

- Order flow test
- Payment gateway validation
- Inventory sync check

B. AUDIT & COMPLIANCE CHECKLISTS**B1. Banking Audit Checklist**

- ✓ RPO/RTO formally approved by business
- ✓ Synchronous AG configured for Tier-0
- ✓ Log backups ≤ 5 minutes
- ✓ Encryption at rest and in transit
- ✓ Quarterly DR test evidence
- ✓ Immutable backups enabled
- ✓ Separation of duties enforced

B2. Healthcare Audit Checklist

- ✓ HIPAA-compliant encryption
- ✓ Access logging enabled
- ✓ Manual failover approvals documented
- ✓ DR testing outside clinical hours
- ✓ Restore integrity validation documented

B3. Retail Audit Checklist

- ✓ Seasonal DR scaling planned
- ✓ Peak-period DR approvals documented
- ✓ Backup retention aligned with policy
- ✓ Cost optimization reviewed annually

C. SLA WORDING (CONTRACT-READY)**C1. Banking SLA Example**

The Service Provider shall ensure:

- **RPO:** Maximum data loss shall not exceed **1 minute**
- **RTO:** Service restoration shall occur within **15 minutes**
- Failure to meet these objectives constitutes a **Severity-1 breach**

C2. Healthcare SLA Example

The Service Provider shall ensure:

- **RPO:** Patient data loss shall not exceed **5 minutes**
- **RTO:** Clinical systems shall be restored within **30 minutes**
- All recovery actions must preserve data integrity and auditability

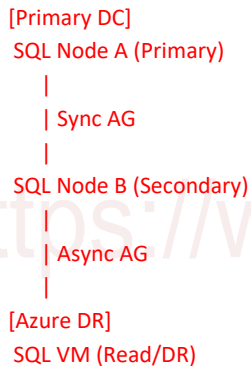
C3. Retail SLA Example

The Service Provider shall ensure:

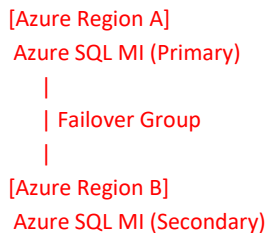
- **RPO:** Transaction data loss shall not exceed **10 minutes**
- **RTO:** Customer-facing systems shall be restored within **30 minutes**
- Peak season SLAs override standard operating SLAs

D. ARCHITECTURE DIAGRAMS

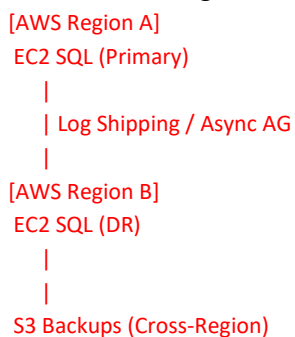
D1. Banking – On-Prem + Azure Hybrid



D2. Healthcare – Azure Managed Instance



D3. Retail – AWS Multi-Region



E. FINAL GOVERNANCE MODEL

Role	Responsibility
DBA	DR setup, testing, execution
Infra / Cloud	Platform availability
Security	Compliance, encryption
Business Owner	RPO/RTO ownership
Audit	Evidence validation

KEY EXECUTIVE SUMMARY

- **Banking:** Zero data loss > cost
- **Healthcare:** Data integrity > automation
- **Retail:** Revenue continuity > cost optimization

RPO/RTO define architecture, not the other way around.

<https://www.sqldbachamps.com/>

1. DR TESTING TEMPLATES – NOTES & EXECUTION RECORD

This template is designed to **prove RPO/RTO compliance**, not just “that DR works” (which auditors often reject).

1.1 DR Test Overview

DR Test ID: DR-YYYY-XXX

Test Type:

- ☐ Backup Restore
- ☐ AG Failover
- ☐ Log Shipping Promotion
- ☐ Full DR Simulation

Environment:

- ☐ On-Prem
- ☐ Azure
- ☐ AWS

SQL Server Version: 2016 / 2017 / 2019 / 2022 / 2025

Workload Type:

- ☐ OLTP
- ☐ OLAP
- ☐ Hybrid

Industry Context:

- ☐ Banking
- ☐ Healthcare
- ☐ Retail

1.2 Business Objectives

Objective	Target
Approved RPO	___ minutes
Approved RTO	___ minutes
Business Owner Approval	☐ Yes / ☐ No

Notes

- Banking: Must include regulator-aligned thresholds
- Healthcare: Must include patient safety consideration
- Retail: Must consider peak/seasonal overrides

1.3 Pre-Test Checklist (Mandatory)

- ✓ Change window approved
- ✓ Stakeholders notified
- ✓ Backups verified successful
- ✓ Replication/AG healthy
- ✓ Rollback plan documented

Healthcare Note:

DR tests must not interfere with clinical operations.

1.4 Test Execution Timeline

Step	Description	Start Time	End Time
Failure Simulation	e.g., SQL service stopped		

Step	Description	Start Time	End Time
DR Trigger	Failover / Restore started		
Database Online	DB accessible		
App Validation	App connectivity verified		
Business Sign-off	Accepted		

1.5 RPO Validation

Metric	Value
Last Transaction Timestamp	
Restore Point Used	
Data Loss Calculated	___ minutes
RPO Met?	☐ Yes / ☐ No

Banking: Any RPO breach = Severity-1

Healthcare: Validate patient record completeness

Retail: Validate order/payment gaps

1.6 RTO Validation

Metric	Value
Approved RTO	___ minutes
Actual RTO	___ minutes
RTO Met?	☐ Yes / ☐ No

1.7 Post-Test Validation

- ✓ Data consistency checks passed
- ✓ SQL Agent jobs validated
- ✓ Application transactions tested
- ✓ Monitoring alerts restored

1.8 Issues & Remediation

Issue	Impact	Owner	Target Date
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1.9 Final Sign-Off

Role	Name	Signature	Date
DBA Lead			
Business Owner			
Compliance (if required)			

2. INCIDENT REPORT TEMPLATE (DR / OUTAGE / DATA LOSS)

This template aligns with ITIL, ISO 22301, SOX, HIPAA, and common regulator expectations.

2.1 Incident Identification

Incident ID: INC-YYYY-XXX

Date & Time Detected:

Environment: On-Prem / Azure / AWS

SQL Server Version:

Affected System(s):

2.2 Incident Classification

Category	Selection
Incident Type	☐ Outage ☐ Data Corruption ☐ Security ☐ Performance
Severity	☐ Sev-1 ☐ Sev-2 ☐ Sev-3
Workload	☐ OLTP ☐ OLAP
Industry	☐ Banking ☐ Healthcare ☐ Retail

2.3 Business Impact Assessment

Area	Impact
Data Loss	☐ None ☐ Within RPO ☐ RPO Breach
Downtime	___ minutes
Customers Affected	
Financial Impact	
Regulatory Impact	☐ Yes / ☐ No

Healthcare Mandatory:

Was patient care affected? ☐ Yes / ☐ No

2.4 Timeline of Events

Time Event

Issue detected

DBA engaged

DR initiated

Service restored

2.5 Root Cause Analysis

Primary Cause:

- ☐ Hardware
- ☐ Network
- ☐ Storage
- ☐ SQL Server
- ☐ Human Error
- ☐ Cybersecurity

Contributing Factors:

- Configuration gap
- Monitoring failure
- Capacity issue

2.6 Recovery Actions Taken

Action	Owner	Time
Failover / Restore		

Action	Owner	Time
Validation		
Business Confirmation		

2.7 RPO / RTO Outcome

Metric	Result
Approved RPO	
Actual RPO	
Approved RTO	
Actual RTO	
SLA Breach	☐ Yes / ☐ No

2.8 Compliance & Notification

Requirement	Status
Regulator Notified	☐ Yes / ☐ No
Customer Notification	☐ Yes / ☐ No
Security Review	☐ Yes / ☐ No

Banking: Regulator notification timeline recorded

Healthcare: HIPAA breach assessment completed

Retail: Customer impact communication documented

2.9 Corrective & Preventive Actions (CAPA)

Action Owner Due Date

2.10 Closure & Approval

Role	Name	Date
DBA Lead		
Business Owner		
Compliance / Security		

3. EXECUTIVE NOTES (Why These Templates Matter)

- Auditors **do not accept screenshots alone**
- RPO/RTO must be **measured, not claimed**
- Incident records must show **decision-making, not just recovery**
- Industry context **changes approval and notification flow**

1. DR Testing Template

Title: Disaster Recovery (DR) Test Template

1. Test Details

- Test Name:
- Date:
- Test Coordinator:
- Systems/Services Tested:
- Test Objective:

2. Scope

- In-Scope Systems:
- Out-of-Scope Systems:
- Dependencies:

3. Test Scenarios

Scenario ID	Description	Steps	Expected Outcome	Actual Outcome	Status (Pass/Fail)	Comments
1						
2						

4. Test Results Summary

- Overall Status:
- Issues Identified:
- Recommendations:

5. Sign-Off

- Test Coordinator:
- Manager Approval:

2. Incident Report Template

Title: Incident Report

1. Incident Details

- Incident ID:
- Date & Time Detected:
- Reported By:
- Incident Type:

2. Description of Incident

- Summary:
- Systems Affected:
- Impact Assessment:

3. Root Cause Analysis

- Immediate Cause:
- Underlying Cause:

4. Resolution & Recovery

- Actions Taken:
- Recovery Steps:
- Time to Resolution:

5. Preventive Measures

- Recommendations:
- Responsible Person/Team:

6. Sign-Off

- Incident Manager:
- Approver:

3. Banking – DR Test Example

Title: Banking – DR Test Example

1. Test Details

- **Test Name:** Core Banking System DR Test
- **Date:** 2025-12-10
- **Test Coordinator:** John Doe
- **Systems/Services Tested:** Core Banking, Online Banking, ATM Network
- **Test Objective:** Verify recovery of core banking systems in case of data center outage

2. Scope

- **In-Scope Systems:** Core Banking, Online Banking
- **Out-of-Scope Systems:** Non-critical internal apps
- **Dependencies:** Network connectivity, backup storage

3. Test Scenarios

Scenario ID	Description	Steps	Expected Outcome	Actual Outcome	Status	Comments
1	Simulate DC outage	Power down primary DC	DR site activates	DR site activated	Pass	None
2	Restore core banking DB	Restore latest backup	DB operational	DB restored successfully	Pass	None

4. Test Results Summary

- **Overall Status:** Pass
- **Issues Identified:** None
- **Recommendations:** Regular quarterly DR testing

5. Sign-Off

- **Test Coordinator:** John Doe
- **Manager Approval:** Jane Smith

4. Healthcare – Incident Example

Title: Healthcare Incident Report Example

1. Incident Details

- **Incident ID:** HC-2025-001
- **Date & Time Detected:** 2025-11-15 09:30
- **Reported By:** Nurse Jane Lee
- **Incident Type:** Patient Data Access Failure

2. Description of Incident

- **Summary:** Staff unable to access patient records during system maintenance
- **Systems Affected:** Electronic Health Records (EHR)
- **Impact Assessment:** Delayed patient care for 2 hours

3. Root Cause Analysis

- **Immediate Cause:** EHR server failure
- **Underlying Cause:** Incomplete system update

4. Resolution & Recovery

- **Actions Taken:** Rebooted servers, restored backup
- **Recovery Steps:** Verified all patient records accessible
- **Time to Resolution:** 2 hours

5. Preventive Measures

- **Recommendations:** Test updates in staging before production
- **Responsible Person/Team:** IT Operations

6. Sign-Off

- **Incident Manager:** Dr. Alice Morgan
- **Approver:** CIO

5. Retail – DR Event Example

Title: Retail – DR Event Example

1. Event Details

- **Event Name:** POS System Outage
- **Date:** 2025-10-05
- **Coordinator:** Michael Brown
- **Systems Affected:** Point-of-Sale (POS), Inventory Management

2. Description

- **Event Summary:** POS system went offline due to network outage
- **Impact:** 12 stores temporarily unable to process transactions

3. Recovery Actions

Action ID	Description	Responsible	Status	Comments
1	Switch to backup network	IT Team	Completed	15 min downtime
2	Reboot POS servers	Store IT	Completed	Services restored

4. Lessons Learned

- Need redundant network routing for POS
- Staff training on manual transaction processing

5. Sign-Off

- **Event Coordinator:** Michael Brown
- **Manager Approval:** Sarah Wilson

Incident Severity Decision Matrix

(Auditor-Friendly and Business-Aligned)

This matrix removes ambiguity during incidents.

3.1 Severity Matrix

Severity	RPO Breach	RTO Breach	Business Impact	Industry Notes
Sev-1 (Critical)	Any	Any	Financial / Patient / Revenue loss	Mandatory regulator notification (Banking, Healthcare)
Sev-2 (High)	Near breach	Minor	Partial service impact	Management escalation
Sev-3 (Medium)	No	No	Degraded service	Operational handling
Sev-4 (Low)	No	No	No business impact	Informational

3.2 Industry Overrides

Banking

- Any RPO breach = **Sev-1**
- Any data inconsistency = **Sev-1**

Healthcare

- Patient care impact = **Sev-1**
- Data integrity risk = **Sev-1**

Retail

- Peak sales outage > 10 min = **Sev-1**
- Off-peak outage usually Sev-2 or Sev-3

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Mapping DR testing, RPO/RTO evidence, and incident reporting process into ITIL v4 incident and problem management practices. This is structured so that we can implement it directly in your organization while remaining audit-ready.

1. ITIL v4 Concepts Applied

ITIL v4 Concept	How It Maps to Your DR/Incident Process
Incident	Any unplanned service interruption (DB outage, failover, backup failure). Captured in the Incident Report Template.
Problem	The underlying cause of one or more incidents. Root cause analysis (RCA) and corrective actions section of the template.
Event Management	Monitoring alerts for database health, backups, and availability groups. Triggers severity assessment.
Change Enablement	DR test plans, failover operations, or corrective scripts requiring approvals before execution.
Service Level Management (SLM)	RPO/RTO definitions are service targets. Actual performance is measured and reported.
Continual Improvement (CSI)	Lessons learned from incidents and DR tests feed back into process updates.

2. Incident Management Workflow – ITIL v4 Aligned

1. Incident Detection & Logging

- Source: Monitoring alerts, user reports, or DR test failures.
- Log in **Incident Report Template**.
- Capture:
 - Incident ID, timestamp
 - Environment & system affected
 - Severity (use Severity Matrix)
 - Initial impact assessment

2. Categorization & Prioritization

- Categorize by system (e.g., OLTP, EHR, Retail portal)
- Prioritize using RPO/RTO breach and business impact
- Assign severity (Sev-1 → Sev-4)

3. Investigation & Diagnosis

- Run SQL & PowerShell evidence scripts
- Capture last backup, failover state, restore duration
- Determine if this is:
 - A single incident
 - Part of a recurring issue (problem candidate)

4. Resolution & Recovery

- Perform DR action (AG failover, restore backup)
- Record RTO – actual time to service recovery
- Record RPO – maximum data loss duration

5. Incident Closure

- Complete Incident Report Template with:
 - Timeline of events
 - Recovery steps executed
 - RPO/RTO comparison
 - Business sign-off

3. Problem Management Workflow – ITIL v4 Aligned

1. Problem Detection

- Triggered by recurring incidents or post-incident review
- Capture in the **Root Cause Analysis** section of the Incident Report

2. Problem Logging & Categorization

- Assign unique Problem ID
- Categorize by system and business impact
- Link related incidents

3. Root Cause Analysis (RCA)

- Identify primary cause (e.g., misconfigured AG, missing backup, network latency)
- Capture contributing factors
- Document evidence (SQL/PowerShell outputs, logs)

4. Problem Resolution & Known Error Record

- Apply permanent fixes
- Update Known Error Database (KEDB)
- If full resolution not immediate, create workaround procedure

5. Problem Closure

- Verify resolution prevents recurrence
- Update metrics: RPO/RTO improvements, downtime reduction
- Feed into **Continual Improvement Register**

4. Integrating DR Testing

• DR Test = Controlled Incident

- Treat DR simulations like Sev-1 incidents
- Follow **full incident workflow**
- Capture all evidence (RPO/RTO, failover duration)
- Feed post-test findings into **problem management**:
 - Document recurring weaknesses
 - Identify process improvement opportunities

5. Evidence & Audit Alignment

ITIL Stage	Evidence Source
Detection & Logging	Monitoring alerts, DR test logs, Incident Report
Diagnosis	SQL queries for last transaction time, backup validation, AG failover status
Recovery	Restore logs, PowerShell failover scripts, downtime timestamps
RCA	Problem Report, root cause section in Incident Report
Continual Improvement	Lessons learned, CAPA items, SLA/RPO/RTO metrics